

Precise asymptotics of the powered Catalan numbers (OEIS A113227)

Positive Bessel spectrum, all fixed orders, and fixed Fourier sectors

A merged research report built from two manuscripts
(batch 77, manuscripts 57 and 56)

Manuscripts dated 1 October 2026; merged 2 October 2026

Abstract

The powered Catalan numbers a_n count permutations avoiding the vincular pattern 1-23-4 (OEIS A113227). This report prints two manuscripts as two Parts. Part I derives a positive discrete moment representation $a_n = \sum_{\lambda} \omega_{\lambda} \lambda^n$ from the known formal continued fraction, identifies its transform with a branch-safe Bessel quotient, locates every large spectral node, and proves $a_n \sim e^{e-2} (2\pi)^{-1/2} r^{-3/2} B_n(e)$ with $w = W(n/e)$, $r = n/w$ and the Touchard polynomial B_n , an expansion to every fixed algebraic order with explicit c_1 , c_2 , a canonical continuous inverse and an exponential-generating-function corollary; this answers the question of precise asymptotics that Elizalde left open in 2006. Part II proves a zero-free complex wedge for the exact Bessel weight, replaces the spectral sum by a lattice sum with relative error $O(e^{-dn})$ for every $d < \log 2$, and decomposes a_n , for every fixed J , into $J + 1$ exact cutoff-defined Fourier sectors with error bounded by the next complex-saddle envelope; each sector has an all-orders expansion with the same rational coefficients. No convergence of the correction series, growing number of sectors, Borel summation, transseries or exponentially accurate inverse is claimed. The report is unrefereed, and none of it is formalized.

Contents

Guide to this report	2
I Precise asymptotics of the powered Catalan numbers	8
1 Main results and known starting points	8
2 The formal fraction and positive finite measures	9
2.1 A reversible path derivation	9
2.2 Finite fractions	10
2.3 Passage to a unique measure	10
3 A branch safe Bessel representation	11
3.1 Identification on the negative real axis	11
4 Uniform estimates and the large spectral nodes	12
5 The residue expansion to arbitrary order	14

6	Global moment reduction and the saddle expansion	15
6.1	Removing the node displacements	15
6.2	Uniform localization and remainders	16
6.3	An explicit algorithm for every coefficient	16
6.4	The Bell polynomial form	17
7	Canonical continuous inversion	17
7.1	The increasing branch	17
7.2	Controlled inversion of the full expansion	18
7.3	An explicit inverse based on the exact leading core	18
8	The exponential generating function on the positive axis	19
9	Numerical checks and reproducibility	20
10	Literature scope and further questions	21
II	Fixed Fourier sectors for the powered Catalan numbers	21
11	Definitions and the fixed sector theorem	22
12	Global exponentially accurate spectral replacement	23
13	The complex Bessel weight	24
13.1	Schl�fli representation and contour rotation	25
13.2	Uniform Gamma normalization and tails	25
13.3	Finite Gamma sums and a summable product remainder	26
13.4	Zero-freeness and analytic remainder bounds	27
14	Poisson summation and the entire omitted Fourier tail	27
14.1	The real window	27
14.2	One common contour height	28
14.3	Paired vertical endpoints	28
14.4	A global horizontal Gaussian bound	29
15	The all orders expansion of each exact sector	29
16	Ordering the envelopes	30
17	Diagnostics and scope	31

Guide to this report

This guide was written when the two manuscripts were merged. Everything after it, from Part I on, is the manuscripts’ text, except for the passages marked [write, 2 October 2026] and the changes listed under “Conventions of the merge” below.

What the report proves

- **Part I** (source 57). Lemma 2.1 and Section 2.3: every finite truncation of the continued fraction is a probability moment transform, and the limit measure μ is unique, with $a_n = \int s^n d\mu(s)$. Theorem 3.1: the transform is $1 - E/D$ with entire E , D , all zeros of D are positive and simple, and they carry positive weights ω_λ . Proposition 4.2: one node $\lambda_k = k + O(e^{-ck \log k})$ near every large integer, with an exact weight formula. Theorem 1.1: the equivalent (2) and the expansion (3) to every fixed order, with c_1 , c_2 and the algorithm (42). Theorem 7.1 and (50)–(51): the canonical continuous inverse. Corollary 8.1: the exponential generating function is entire, with an all-orders expansion on the positive axis.
- **Part II** (source 56). Lemma 13.1: the exact Bessel weight $Q(z) = 1/(\pi^2 z b(z)^2)$ has an asymptotic expansion, and b is eventually zero-free, in every closed wedge $|\arg z| \leq \theta < \pi/2$. Proposition 12.1: global spectral replacement with relative error $O_d(e^{-dn})$, $d < \log 2$. Theorem 11.1: the fixed-sector decomposition (56), the all-orders expansion (57) of every fixed exact sector at the complex saddle $w_j = W_{-j}(n/e)$, and the envelope ordering (58).

Sources and provenance

The report was built from two manuscripts, both dated 1 October 2026, both delivered in the arrival commit 096ee7b87 of 2 October 2026 as batch 77, manuscripts 57 and 56, and placed in this directory by commit aa7345800 (batch 77P2). Neither pins a ProveIt commit or names a repository path. Their author lines are “A proof and reproducible research report” (source 57) and “A research addendum with reproducible diagnostics” (source 56); no tool or person is named.

Part	Source	Archive and main file	Pages	Labels	Prefix
I	57	oeis-powered-catalan-report.zip, powered-catalan-asymptotics.tex (635 lines)	14	59	pcn:
II	56	oeis-powered-catalan-fourier-addendum.zip, powered-catalan-fourier-addendum.tex (497 lines)	10	32	pcn:fs:

Nesting. Source 56 is an addendum of source 57 and is not self-contained: it cites source 57 (bibliography key `Core`) for the spectrum, the real coefficient algorithm and the interpolation, and its archive embeds source 57’s manuscript, PDF and verification record byte for byte (as `core-report.tex`, `core-report.pdf`, `core-mathematical-verification.md`), together with six byte-identical support files (`residue_expansion.py`, `saddle_expansion.py`, their three outputs and `requirements.txt`). The placement staged each file once, from source 57’s archive. Source 56 does not repeat any proof of source 57.

Where the merge had to choose. (i) The Parts follow the dependency order 57, 56. (ii) Both sources are printed in full; nothing of either is omitted except the title blocks and the two bibliographies, which are merged below. (iii) Source 56 restates the inputs it takes from source 57 (the weight and node estimates (59)–(60), the coefficients c_0 , c_1 , c_2 , α_0 , α_1 , α_2 , the generators (70), (71) and (81)). These restatements are kept, because Part II continues them into the complex wedge and its proofs cite them by its own labels; each now names the place in Part I where the input is proved. (iv) Three symbols of source 56 were renamed to avoid collisions with Part I; see the notation table.

How the Parts depend on one another

Part II imports from Part I the positive discrete spectrum (Theorem 3.1), the node and weight estimates (Proposition 4.2), the real coefficient algorithm (42) and the interpolation (43). It proves

everything else itself. The sector expansion (57) at $j = 0$ is the all-orders expansion (3) of Part I for the exact principal sector $I_0(n)$, whereas Part I expands a_n itself. The two statements differ by the nonzero sectors, which are smaller than every algebraic order (Theorem 11.1), so they are consistent. Part II does not change the inverse of Part I and proves no exponentially accurate inverse.

Relation to other reports

[write, 2 October 2026] *Sibling report on the pattern 12-34*. Batch 77 also opened the report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a113226-vincular-avoiders`, on permutations avoiding the vincular pattern 12-34 (OEIS A113226). Both reports prove precise asymptotics for a length-four vincular pattern whose asymptotics Elizalde studied in 2006 [2], and both cite Callan’s bijection paper on 1-23-4 [3]. Otherwise they share no theorem and no method: the A113226 report works from a closed exponential generating function (an inverse-square-root singularity and a stretched-exponential saddle in powers of $n^{-1/3}$), this report from a positive Bessel spectrum and a Touchard saddle. The two sequences are different, as Section 1 says. They were therefore kept as two reports, by the coordinator’s decision at the write, rather than printing these two manuscripts as further Parts of the A113226 report.

The Touchard saddle $w = W(n/e)$ of this report is the analogue, at parameter $\theta = e$, of the Lambert saddle $r = W_0(n)$ of the Bell numbers ($\theta = 1$). The Bell numbers are treated in the neighbouring report `a277364-bell-asymptotics` of this collection directory and in the chapter “The Bell numbers: a Lambert saddle” of the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/` (label `q2:sec:bell`; all-orders theorem `q2:thm:bell`). Remark `q2:rem:bell-sectors` of that volume keeps the nonprincipal complex saddles of the Bell numbers *formal*. Part II proves fixed complex sectors rigorously, but for A113227, a different sequence with an exact Bessel weight; it does not upgrade that remark (see the note at the end of Part II).

Placement

Part II is the batch’s closest case to the repository’s rule that transseries material goes to `Analysis/Transseries/`. It was kept in this collection, beside its foundation, because it proves a finite decomposition into a fixed number of exact cutoff-defined Fourier sectors with algebraic expansions and constructs no transseries, Borel sum or Stokes data; its own scope paragraph (Section 17) says so. If Vladimir prefers every exponentially ordered sector theorem in the transseries tree, Part II can be filed there whole and replaced here by a cross-reference.

Notation

Each Part keeps its source’s letters, except for the three renamed symbols marked below. Symbols not in the table are local to the Part that uses them. The table prints the tempting false reading beside the true one.

Symbol	Meaning and where	Do not read it as
a_n	The count (both Parts); in Part II, for real n , the interpolation $\mathcal{A}(n)$ of (43).	Part I’s Bessel combination $a(t)$ (16); Part II’s window endpoint $a = \lfloor r/8 \rfloor + \frac{1}{2}$.
w, r	$w = W(n/e)$ (real branch), $r = n/w = e^{w+1}$ (both Parts).	

Symbol	Meaning and where	Do not read it as
w_j, t_j	Part II: $w_j = W_{-j}(n/e)$, $t_j = n/w_j$, the complex saddles; $w_0 = w$, $t_0 = r$.	Part I's w_0, r_0, N_0 in (48) solve the inverse core equation $e^{w_0+1}(w_0^2+1) = L$; that w_0 is <i>not</i> $W(n/e)$.
M_n, M_j, E_j	M_n of (1) (Part I) is M_0 of Part II; M_j is the complex leading factor at t_j and $E_j = M_j $ the envelope.	E_j is not Part I's entire function $E(t)$ (12), nor the exponential generating function $\mathcal{E}(z)$.
$E_S(z)$	Part II: the remainder in Schläfli's representation (64). <i>Renamed</i> from source 56's $E(z)$.	Not $E(t)$ of (12).
$B_n(\theta)$	Touchard (Bell) polynomial (Part I); $B_n(e)$ is not the Bell number.	Part II's upper window endpoint $B = \lfloor 4r \rfloor + \frac{1}{2}$.
β_{2m}	Bernoulli numbers (both Parts). Part II's generator (71) writes β_{2q} ; <i>renamed</i> from source 56's B_{2q} .	Not a coefficient of $\mathcal{B}$.
$\mathcal{B}(z), \mathcal{B}_m$	The formal series (30) = (70), and its coefficients $\mathcal{B}_m = [s^m]\mathcal{B}(s)$: $1, -\frac{1}{2}, -\frac{5}{24}, -\frac{7}{16}, \dots$, see (32). Part II's $\mathcal{B}_m$ is <i>renamed</i> from source 56's β_m .	$\mathcal{B}_2 = -\frac{5}{24}$ is not the Bernoulli number $\beta_2 = \frac{1}{6}$.
α_ℓ	Coefficients of the weight expansion, (34) = (71) (both Parts; Part II indexes them by m).	
c_j	The rational correction functions (Part I: $c_j(w)$; Part II: $c_\ell(u)$, the same functions).	The constant $0 < c < 1$ of (24) and (59).
$q(t), Q(z)$	Part I: $q = a/b$, a small ratio. Part II: $Q(z) = 1/(\pi^2 z b(z)^2)$, the exact weight, so $\omega_k = Q(\lambda_k)(\dots)$, cf. (26).	Q is not a multiple of q .
J, L	Part I: J is the algebraic order in (3) and of H_J ; L is a log-target in Section 7. Part II: J is the number of retained nonzero Fourier sectors, L the algebraic order of each sector.	Part II's J is not an algebraic order.
H, h	Part I: $H(t)$, the Bessel denominator (18); $H_J(x)$, the inverse (46). Part II: $H = \lfloor R^{1/4} \rfloor$, a cutoff; H_j , horizontal contour integrals; $\mathcal{H}(c) = c - 1 - \log c$; $h(x) = x - 1 - \log x$; $h = \Im t_m$, a contour height.	
$I_j(n), \mathcal{I}(z)$	Part II: exact Fourier sectors (54); the rotated Schläfli integral (66).	Not the modified Bessel function I_ν of Part I, Section 3.
σ	Part I: $\sqrt{r/(w+1)}$. Part II: $\sqrt{r/w}$.	The two widths differ.
d, K, R	Part I: horizontal weights d_h , the inverse constant d_0 ; a Poisson variable K ; $R_k(t)$, and $R_n = a_n/M_n$. Part II: a decay rate $d < \log 2$; a large fixed integer K ; $R = z $.	
P, F, θ	Part I: $P(t) = t^{t/2}/\Gamma(t+1)$, the transform $F(t)$, $F_0(x)$, the Touchard parameter θ . Part II: reciprocal products $P_{j,\delta}$, the tilted exponent F_p , a wedge angle θ .	

Status, scope and non-claims

Neither manuscript states how it was produced; both reached the repository through its incoming channel for AI-assisted research deliveries and are treated as AI-assisted. Both are unrefereed: source 57's verification record calls itself “not a proof-assistant verification or an external peer-review certification”, and source 56 calls its review “component cross-review plus integrated mathematical

review, not formal verification or conventional peer review”. Their computations are high-precision diagnostics, not interval certificates. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts, is as follows.

- No convergence of the infinite correction series (Theorem 1.1, Part II Section 17); the all-orders result is a Poincaré expansion, not an exponentially complete transseries.
- Fixed truncation only: the number of sectors and the algebraic order are fixed; no growing- J theorem, no summation of infinitely many sector expansions, no canonical Borel sum. A fixed algebraic truncation of the principal sector has an error larger than every nonzero Fourier envelope (Remark 11.2).
- The error scales are proved without explicit constants. A numerical Newton residual controls only the root of the truncated inverse equation; a certified enclosure of the true inverse needs an explicit remainder bound. No exponentially accurate inverse is claimed. A finite expansion in inverse logarithms is not claimed to give integer accuracy uniformly.
- The numerical tables are diagnostics. Part II’s quadratures integrate central saddle segments and do not evaluate complete cutoff-defined sectors.
- The literature checks are scoped and do not establish worldwide priority or repository-wide nonduplication. The recurrence, path models and continued fraction are prior art (Callan, Beaton et al., Gladkovskii in OEIS); Poisson summation, complex Gaussian expansion and Lambert- W saddles in Touchard asymptotics are standard (Paris).

Instances of results elsewhere in the repository

Several steps are instances of results proved in the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. No novelty is claimed for these mechanics.

- The saddle $w = W(n/e)$ solves $w + \log w = \log(n/e)$, the linear-logarithmic equation $aX + b \log X = L$ with $a = b = 1$; the volume’s Lambert core, theorem `p0:thm:lambert-core`, solves it on every branch. Part II’s complex saddles $w_j = W_{-j}(n/e)$, with $w_j + \text{Log } w_j = \log(n/e) - 2\pi i j$, are the branches $k = -j$ of the same solution.
- Theorem 7.1 and the explicit inverses (50)–(51) revert a perturbed core equation, as in theorem `p0:thm:perturbed-inversion`. The threshold rule $\min\{n : a_n \geq Y\} = \lceil x_*(\log Y) \rceil$, with a comparison of neighbouring exact values when an enclosure straddles an integer, is the separation step of theorem `p0:thm:staircase`, whose separation inequality is formalized in Lean as `staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. Nothing specific to A113227 is formalized.

Conventions of the merge

- Every label carries the report prefix: `pcn:` in Part I and `pcn:fs:` in Part II, for example `pcn:fs:thm:main`. Guide labels are `pcn:guide:`, Part labels `pcn:part:`, and the few section labels added at the merge `pcn:sec:` and `pcn:fs:sec:`.
- Sections are numbered through the report: source 57’s Section k is Section k here (Sections 1–10), source 56’s is Section $k + 10$ (Sections 11–17). Theorems are numbered within sections,

as in both sources, so source 56’s Theorem 1.1 is Theorem 11.1. Equations are numbered through the report.

- Source 56’s four citations of source 57 (`\cite{Core}`) and its two further mentions of it (“the audited spectrum”, “the companion report”) are now references to Part I, and the bibliography entry `Core`, which described `core-report.pdf` and `core-report.tex` as included in the package, is dropped. The two `DLMF` entries are merged into one.
- Sentences about “the reproducibility package”, “the reproducibility archive”, “the accompanying output” or “the supplied symbolic generator” describe the delivered archives, which survive in commit `096ee7b87`. Which of their files this report ships, and under which names, is listed in the `README`. The delivered PDFs, checksum manifests and source 56’s embedded copies are not shipped.
- Each Part opens with its source’s abstract as delivered.

Part I

Precise asymptotics of the powered Catalan numbers

Source. Batch-77 manuscript 57, archive `oeis-powered-catalan-report.zip`, main file `powered-catalan-asymptotics.tex` (635 lines, 14-page PDF), dated 1 October 2026. Author line: “A proof and reproducible research report”. No pin. Printed in full, with labels prefixed `pcn:`; four section labels and the `[write]` notes were added at the merge.

Abstract (as delivered). The powered Catalan numbers count permutations avoiding the vincular pattern 1-23-4 and form OEIS A113227. We derive a positive discrete moment representation from their known formal continued fraction, identify its transform by a branch-safe Bessel quotient, and determine every sufficiently large spectral node and its weight. This proves a sharp equivalent involving the Bell polynomial $B_n(e)$, including its explicit constant, and gives an expansion to every fixed algebraic order. An exact coefficient algorithm, two displayed corrections, and a canonical continuous inverse accompany the proof. The analytic arguments control the full spectrum and all omitted tails; numerical computations are used only as reproducible diagnostics.

1 Main results and known starting points

Let a_n count permutations $\pi_1 \cdots \pi_n$ for which there are no indices $i < j < j+1 < k$ with

$$\pi_i < \pi_j < \pi_{j+1} < \pi_k.$$

Thus a_n is OEIS A113227, with

$$(a_n)_{n \geq 0} = 1, 1, 2, 6, 23, 105, 549, 3207, 20577, 143239, \dots$$

The adjacency of the middle pair is essential. This sequence is different from avoidance of the classical pattern 1234, fully consecutive 1234, or the vincular pattern 12-34.

For $n > 0$, throughout the report put

$$w = W(n/e), \quad r = \frac{n}{w} = e^{w+1}, \quad M_n = \frac{e^{-2} r^{n-3/2} \exp(r-n)}{\sqrt{2\pi(w+1)}}, \quad (1)$$

where W is the positive real Lambert function. Write

$$B_n(\theta) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \theta^k$$

for the Bell or Touchard polynomial; $B_n(e)$ is not the ordinary Bell number.

Theorem 1.1. *The powered Catalan numbers satisfy*

$$a_n \sim M_n \sim \frac{e^{e-2}}{\sqrt{2\pi}} r^{-3/2} B_n(e). \quad (2)$$

For every fixed integer $J \geq 1$,

$$a_n = M_n \left(\sum_{j=0}^{J-1} \frac{c_j(w)}{r^j} + O_J(r^{-J}) \right), \quad (3)$$

where $c_0 = 1$ and the c_j are explicitly computable rational functions. Their denominators divide a constant times $(w + 1)^{3j}$, and their numerators have degree at most $3j$. In particular,

$$c_1(w) = \frac{20w^3 + 86w^2 + 128w + 67}{24(w + 1)^3}, \quad (4)$$

$$c_2(w) = \frac{1168w^6 + 9968w^5 + 34692w^4 + 64536w^3 + 69380w^2 + 42208w + 11857}{1152(w + 1)^6}. \quad (5)$$

The constants in (3) may depend on J ; no convergence of the infinite correction series is asserted.

The recurrence, combinatorial models, and formal fraction used below are known. Callan [3] gives the recurrence and valley-marked Dyck paths; Beaton et al. [4] develop the powered-Catalan terminology and further models. OEIS credits the formal continued fraction to Sergei N. Gladkovskii, 10 October 2012 [1]. Elizalde [2] established asymptotic bounds and explicitly left a precise asymptotic for this pattern open. The contribution proved here is the sharp equivalent, the global positive spectral representation supporting it, and the controlled refinements. A scoped literature review is summarized in Section 10; it does not certify worldwide priority or repository-wide nonduplication.

[write, 2 October 2026] Part II (source 56) refines Theorem 1.1 beyond all algebraic orders. For every fixed J it writes a_n as the sum of $J + 1$ exact cutoff-defined Fourier sectors, with an error bounded by the next complex-saddle envelope E_{J+1} , and each fixed sector has an all-orders expansion with the same rational functions c_j , evaluated at the complex saddle $w_j = W_{-j}(n/e)$ (Theorem 11.1). The nonzero sectors are of relative size $\exp(-2\pi^2 j^2 r/(w + 1) + \dots)$, so they do not affect (3).

2 The formal fraction and positive finite measures

2.1 A reversible path derivation

In the known path model, a Dyck path of semilength n has $h + 1$ possible marks at every valley of height h . Keep mark-zero valleys unchanged. Replace a valley DU with a nonzero mark by a long horizontal step at height $h + 1$, with h possible colors. Distinct valleys do not overlap, and expansion of each horizontal step to DU reverses the operation. The resulting Schröder paths have unit up/down weights and horizontal weights

$$d_0 = 0, \quad d_h = h - 1 \quad (h \geq 1).$$

First-return decomposition therefore gives, as a formal series,

$$A(x) = \sum_{n \geq 0} a_n x^n = 1 + \frac{x}{U_0(x) - x}, \quad U_k(x) = 1 - kx - \frac{x}{U_{k+1}(x)} \quad (k \geq 0). \quad (6)$$

Equivalently it is the Thron fraction with successive denominator weights $0, 0, 1, 2, \dots$ and unit numerators. This also falls under general nonnegative Thron-fraction moment-positivity results [10]; the following direct proof is sufficient here. Nothing in the argument treats the divergent series $A(x)$ as analytic near zero.

2.2 Finite fractions

Truncate (6) at $U_m = 1 - mx$, with $m \geq 1$, and denote the resulting rational function by A_m . For $k \geq 1$ let $R_k(t) = tU_k(1/t)$ and at the root let $R_0 = tU_0(1/t) - 1$. Then

$$R_m = t - m, \quad R_k = t - k - \frac{t}{R_{k+1}} \quad (1 \leq k < m), \quad R_0 = t - 1 - \frac{t}{R_1}, \quad A_m(1/t) = 1 + \frac{1}{R_0}. \quad (7)$$

The exceptional root shift must be retained.

Lemma 2.1. *Each $A_m(1/t)$ is a probability moment transform*

$$A_m(1/t) = \sum_{j=1}^{N_m} \omega_{m,j} \frac{t}{t - \lambda_{m,j}}, \quad \omega_{m,j} > 0, \quad \lambda_{m,j} > 0, \quad \sum_j \omega_{m,j} = 1.$$

Proof. Consider a real rational function

$$R(t) = t - b - \sum_j \frac{c_j}{t - \eta_j}, \quad c_j > 0, \quad \eta_j > 0, \quad R(0) < 0.$$

Repeated poles can be combined. Since $R'(t) = 1 + \sum_j c_j / (t - \eta_j)^2 > 0$, its zeros are simple and interlace its poles. There is no nonpositive zero, one zero between zero and its first pole, one between each successive pair, and one after its last pole. Its reciprocal consequently has the partial fraction expansion

$$\frac{1}{R(t)} = \sum_j \frac{\rho_j}{t - \xi_j}, \quad \rho_j = \frac{1}{R'(\xi_j)} > 0, \quad \sum_j \rho_j = 1.$$

The sum of residues follows from $1/R(t) \sim 1/t$ at infinity. Hence

$$\frac{t}{R(t)} = 1 + \sum_j \frac{\xi_j \rho_j}{t - \xi_j}.$$

This proves preservation of the stated form under (7), starting from $t - m$. Also $R_k(0) = -k$ for $k \geq 1$ and $R_0(0) = -1$. At the root evaluation at zero gives $\sum_j \rho_j / \xi_j = 1$, and

$$1 + \frac{1}{R_0(t)} = \sum_j \frac{\rho_j}{\xi_j} \frac{t}{t - \xi_j}.$$

These are the asserted probability weights. □

2.3 Passage to a unique measure

Every fixed formal coefficient of A_m eventually stabilizes to that of A . The associated probability measures μ_m have bounded first moments, so they are tight. Along any weakly convergent subsequence, a bound on the next moment gives uniform integrability of each fixed power. The limit μ thus has moments

$$a_n = \int_0^\infty s^n d\mu(s), \quad n \geq 0. \quad (8)$$

It is unique: the permutation interpretation gives $a_n \leq n!$, so monotone convergence implies

$$\int \exp(\delta s) d\mu(s) = \sum_{n \geq 0} \frac{\delta^n a_n}{n!} \leq \frac{1}{1 - \delta} \quad (0 < \delta < 1).$$

The moments determine the moment-generating function near zero and therefore determine the measure. Thus the whole sequence μ_m converges to μ . In particular the function

$$F(t) = \int_0^\infty \frac{t}{t-s} d\mu(s), \quad t \in \mathbb{C} \setminus [0, \infty), \quad (9)$$

is the limit of the finite fractions. It is this analytic transform, rather than the divergent ordinary series, that will be identified.

3 A branch safe Bessel representation

3.1 Identification on the negative real axis

Fix $t = -s < 0$ and define

$$V_k = \frac{I_{k+s-1}(2\sqrt{s})}{\sqrt{s} I_{k+s}(2\sqrt{s})}.$$

All these quantities are positive: denominator orders are positive and numerator orders exceed -1 . The modified Bessel recurrence gives

$$V_k = 1 + \frac{k}{s} + \frac{1/s}{V_{k+1}}, \quad V_k \geq 1 + \frac{k}{s}. \quad (10)$$

The truncated $U_k(-1/s)$ satisfy the same recurrence, with terminal value $1 + m/s$. Its discrepancy from V_m is at most $(1/s)/(1 + (m+1)/s)$. Each preceding step contracts the discrepancy by at most

$$\frac{1/s}{(1 + (k+1)/s)^2} = \frac{s}{(s + k + 1)^2} \leq \frac{1}{4(k+1)}.$$

Their product tends to zero factorially. This proves the required branch selection directly, without an assumption about a minimal solution of an infinite recurrence.

In Bessel notation the resulting transform is

$$F(t) = 1 - \frac{J_{-t}(2\sqrt{t})}{\sqrt{t} J_{-t-1}(2\sqrt{t}) + J_{-t}(2\sqrt{t})}. \quad (11)$$

A formula with no square-root or logarithmic branch convention is preferable for global use. Define

$$E(t) = \sum_{j \geq 0} \frac{(-t)^j}{j! \Gamma(j-t+1)}, \quad (12)$$

$$D(t) = \sum_{j \geq 0} \frac{(j-t+1)(-t)^j}{j! \Gamma(j-t+1)}. \quad (13)$$

The gamma growth in j proves local uniform convergence for complex t , so both functions are entire. The Bessel series [11, §10.2] identifies (11), after cancellation of its common factor $t^{-t/2}$, with

$$F(t) = 1 - \frac{E(t)}{D(t)}. \quad (14)$$

The negative-axis equality and the identity theorem give its analytic continuation.

Theorem 3.1. *All zeros of D are positive and simple. They are exactly the support points of a probability measure with positive weights, and*

$$F(t) = \sum_{\lambda: D(\lambda)=0} \omega_\lambda \frac{t}{t-\lambda}, \quad a_n = \sum_{\lambda: D(\lambda)=0} \omega_\lambda \lambda^n, \quad \omega_\lambda = -\frac{E(\lambda)}{\lambda D'(\lambda)} > 0. \quad (15)$$

The support has no finite accumulation point and is bounded away from zero.

Proof. The functions E, D have no common zero. At zero both equal 1. At a nonzero common zero, (11) would make two consecutive Bessel functions vanish at the same nonzero argument. Their recurrence would make every forward consecutive order vanish, contradicting the fixed-argument large-positive-order series.

The quotient (14) agrees with the Stieltjes transform (9); hence its genuine poles can occur only on the positive axis. Every such pole is simple, since a Cauchy transform of a finite positive measure has at most a simple meromorphic pole, and its residue is the corresponding positive atom. Between poles the quotient continues analytically with real values. Stieltjes inversion therefore gives zero measure there, ruling out continuous mass. Entire-function zeros are locally finite. Finally $F(0) = 0$ rules out an atom at zero, while $D(0) = 1$ leaves a neighborhood of zero free of support. This proves every assertion, including the absence of an additional entire contribution or a hidden cut. $\square$

4 Uniform estimates and the large spectral nodes

For real $t > 0$ define

$$a(t) = J_t(2\sqrt{t}) - \sqrt{t} J_{t+1}(2\sqrt{t}), \quad (16)$$

$$b(t) = \sqrt{t} Y_{t+1}(2\sqrt{t}) - Y_t(2\sqrt{t}), \quad q(t) = \frac{a(t)}{b(t)}. \quad (17)$$

Order reflection [11, §10.4] writes the denominator in (11) as

$$H(t) = \cos(\pi t) a(t) + \sin(\pi t) b(t). \quad (18)$$

Lemma 4.1. *Let $P(t) = t^{t/2}/\Gamma(t+1)$. As real $t \rightarrow \infty$,*

$$J_t(2\sqrt{t}) = P(t)(e^{-1} + O(t^{-1})), \quad (19)$$

$$b(t) = -\frac{e}{\pi} t \Gamma(t) t^{-t/2} (1 + O(t^{-1})), \quad (20)$$

$$a(t) = O(P(t)), \quad a'(t) = O(P(t) \log t), \quad b'(t) = O(|b(t)| \log t). \quad (21)$$

Consequently, for every fixed $0 < c < 1$ and all sufficiently large t ,

$$|q(t)| + |q'(t)| \leq \exp(-ct \log t). \quad (22)$$

Proof. The absolutely convergent series gives

$$\frac{J_t(2\sqrt{t})}{P(t)} = \sum_{j \geq 0} \frac{(-1)^j}{j!} \prod_{h=1}^j (1 + h/t)^{-1}.$$

The product differs from 1 by at most $j(j+1)/(2t)$, whose sum against $1/j!$ is finite. Its derivative in t is bounded by $j(j+1)/(2t^2)$. Since $P'/P = O(\log t)$, the same reasoning for the adjacent order proves the J and a estimates without differentiating an unspecified remainder.

Schl fli's formula [11, 10.9.7] is

$$Y_\nu(z) = \frac{1}{\pi} \int_0^\pi \sin(z \sin \theta - \nu \theta) d\theta - \frac{1}{\pi} \int_0^\infty (e^{\nu u} + \cos(\pi \nu) e^{-\nu u}) e^{-z \sinh u} du.$$

The bounded integral and the decaying integral contribute $O(\sqrt{t})$ to b and to its total t -derivative. In the growing integral use $v = \sqrt{t} e^u$. This gives the exact dominant integral

$$b(t) = -\frac{t^{-t/2}}{\pi} \int_{\sqrt{t}}^\infty (v-1)v^{t-1} \exp(-v + t/v) dv + O(\sqrt{t}). \quad (23)$$

For V of gamma distribution with shape t and scale 1, its integral normalized by $\Gamma(t)$ is

$$\mathbb{E}[(V-1)e^{t/V} \mathbf{1}_{\{V \geq \sqrt{t}\}}] = et(1 + O(t^{-1})).$$

Indeed on $[t/2, 2t]$ Taylor expansion at $V = t$ uses $\mathbb{E}(V - t) = 0$ and $\text{Var}(V) = t$. On the lower tail $e^{t/V} \leq e^{\sqrt{t}}$, absorbed by the gamma bound $e^{-c_0 t}$; on the upper tail it is at most $e^{1/2}$. This proves (20).

Differentiation of the growing integrand including its prefactor adds

$$\log v - \frac{1}{2} \log t - \frac{1}{2} + \frac{1}{v}.$$

The same gamma localization bounds its weighted integral by $O(\log t)$ times the main integral. The lower-endpoint term is $O(t^{-1/2})$ in absolute size and is negligible relative to that main term. This proves the last derivative bound. Stirling's formula now gives (22). $\square$

Proposition 4.2. *For every sufficiently large integer k there is exactly one spectral node λ_k in $(k - \frac{1}{2}, k + \frac{1}{2})$, there are no other sufficiently large nodes, and*

$$\lambda_k = k + O(\exp(-ck \log k)) \quad (0 < c < 1). \quad (24)$$

At those nodes the weight has the exact expression

$$\omega_k = \frac{1}{\lambda_k b(\lambda_k)^2 \{\pi^2(1 + q(\lambda_k)^2) + \pi q'(\lambda_k)\}}. \quad (25)$$

In particular,

$$\omega_k = \frac{1}{\pi^2 k b(k)^2} (1 + O(\exp(-ck \log k))), \quad (26)$$

$$\omega_k \sim e^{-2} \frac{k^{k-3}}{\Gamma(k)^2} \sim \frac{k^{-2} \exp(-k \log k + 2k)}{2\pi e^2} \sim \frac{e^{-2}}{\sqrt{2\pi}} k^{-3/2} \frac{e^k}{k!}. \quad (27)$$

Proof. For large t , $b(t) < 0$. Zeros of (18) solve $\tan(\pi t) + q(t) = 0$. On each indicated interval the left side ranges from $-\infty$ to $+\infty$ and has derivative at least $\pi - |q'| > 0$. The endpoints are not zeros because $b \neq 0$ there. This proves completeness and (24).

At a pole, reflection gives the residue formula

$$\omega_k = -\frac{J_t + q(t)Y_t}{tb(t)\{\pi(1 + q(t)^2) + q'(t)\}} \Big|_{t=\lambda_k},$$

where the argument of each Bessel function is $2\sqrt{t}$. The Wronskian [11, 10.5.2] implies

$$J_t b + a Y_t = \sqrt{t} (J_t Y_{t+1} - J_{t+1} Y_t) = -1/\pi.$$

This proves (25). Bounds (21) and (22) permit replacing λ_k by k , giving (26). Equations (20) and Stirling's formula finish the proof. $\square$

A further expansion, not needed for completeness of the spectrum, gives

$$a(t) = P(t)e^{-1} \left(t^{-2} + \frac{3}{2}t^{-3} - \frac{97}{24}t^{-4} + O(t^{-5}) \right).$$

It follows that the displacement is eventually positive and

$$\lambda_k - k \sim e^{-2} \frac{k^{k-4}}{\Gamma(k)^2} \sim \frac{\omega_k}{k}. \quad (28)$$

The finite number of remaining low nodes cause no difficulty in large moments.

5 The residue expansion to arbitrary order

Introduce the following *formal* series:

$$T_0(z) = \sum_{j \geq 0} \frac{1}{j!} \prod_{h=0}^{j-1} (1 - hz)^{-1}, \quad (29)$$

$$T_1(z) = \sum_{j \geq 0} \frac{1}{j!} \prod_{h=1}^j (1 - hz)^{-1}, \quad \mathcal{B}(z) = e^{-1} \{T_0(z) - zT_1(z)\}. \quad (30)$$

Each formal coefficient is a convergent Poisson sum of a polynomial in j . The expressions are not infinite convergent sums to be evaluated at $z = 1/k$: individual factors then have poles for large j .

Lemma 5.1. *For each fixed order, the following is an asymptotic expansion with a remainder of the corresponding inverse power:*

$$b(k) \sim -\frac{ek\Gamma(k)k^{-k/2}}{\pi} \mathcal{B}(1/k). \quad (31)$$

Its first coefficients are

$$\mathcal{B}(z) = 1 - \frac{z}{2} - \frac{5z^2}{24} - \frac{7z^3}{16} - \frac{7057z^4}{5760} + O(z^5), \quad (32)$$

$$\mathcal{B}(z)^{-2} = 1 + z + \frac{7}{6}z^2 + 2z^3 + \frac{1739}{360}z^4 + O(z^5). \quad (33)$$

Proof. Use (23) with $t = k$. Fix $0 < \epsilon < 1/2$ and retain terms $0 \leq j \leq m = \lfloor k^\epsilon \rfloor$ in the exponential $\exp(k/v)$. On $[k/2, 2k]$ the discarded remainder is at most a constant times $2^{m+1}/(m+1)!$, smaller than every inverse power of k . The exterior contribution of the original integral is exponentially small as in Lemma 4.1.

For retained j the gamma integrals have shapes $k - j + 1$ and $k - j$. Extending their central intervals to $(0, \infty)$ changes them by uniformly exponentially small relative errors, since their centers differ from k by $O(k^\epsilon)$. The extended integral of the j th term is

$$\frac{k^j}{j!} \{ \Gamma(k - j + 1) - \Gamma(k - j) \}.$$

After normalization by $k\Gamma(k)$, these terms give precisely $T_0(1/k) - k^{-1}T_1(1/k)$, truncated at $j = m$.

For either reciprocal product, expansion through degree J has remainder bounded uniformly for $j \leq m$ by

$$C_J k^{-J-1} (1+j)^{2J+2} \exp(Cj^2/k).$$

This follows by expanding its logarithm or by differentiating the product. The exponential factor is bounded for $\epsilon < 1/2$, and the polynomial factor is summable against $1/j!$. Extending every coefficient sum to all j costs a further superalgebraic factorial tail. Thus the remainder is $O_J(k^{-J-1})$ after retaining degrees through J , proving (31). The coefficients follow by exact polynomial arithmetic. $\square$

Let β_{2m} denote the Bernoulli numbers, to distinguish them from $B_n(\theta)$. Define α_ℓ by

$$\sum_{\ell \geq 0} \alpha_\ell z^\ell = \mathcal{B}(z)^{-2} \exp\left(-\sum_{m \geq 1} \frac{\beta_{2m} z^{2m-1}}{m(2m-1)}\right). \quad (34)$$

Combining (26), Lemma 5.1, and Stirling's formula proves

$$\omega_k = \frac{k^{-2} \exp(-k \log k + 2k)}{2\pi e^2} \left(\sum_{\ell=0}^{J-1} \frac{\alpha_\ell}{k^\ell} + O_J(k^{-J}) \right). \quad (35)$$

In particular

$$\alpha_0 = 1, \quad \alpha_1 = \frac{5}{6}, \quad \alpha_2 = \frac{73}{72}, \quad \alpha_3 = \frac{11821}{6480}, \quad \alpha_4 = \frac{702533}{155520}.$$

For comparison with Bell polynomials the equivalent normalization is

$$\frac{\omega_k}{[e^{-2/\sqrt{2\pi}}]k^{-3/2}e^k/k!} \sim 1 + \frac{11}{12k} + \frac{313}{288k^2} + \frac{98959}{51840k^3} + \dots \quad (36)$$

An exact implementation is included in the reproducibility package. Polynomial moments are evaluated using $e^{-1} \sum_{j \geq 0} j^p/j! = B_p(1)$, so these coefficients are not numerical fits.

6 Global moment reduction and the saddle expansion

6.1 Removing the node displacements

The exact positive moment representation separates into finitely many low nodes and the λ_k of Proposition 4.2. For $k \geq \sqrt{r}$, (24) gives, for every fixed J ,

$$\left(\frac{\lambda_k}{k}\right)^n = 1 + o(r^{-J})$$

uniformly in this range. For nodes indexed by $k < \sqrt{r}$, including all low nodes, positivity and total mass one bound their entire contribution by $(2\sqrt{r})^n$ for large n . This is smaller than every algebraic multiple of a single term near $k = r$, since that term has logarithm $n \log r - n + r + O(\log r)$. Thus the actual nodes can be replaced by integers at every algebraic order.

Consequently (35) reduces the problem to real lattice sums with exponent

$$\phi(t) = n \log t - t \log t + 2t. \quad (37)$$

The unique maximum is at $t = r$, where

$$\phi(r) = n \log r - n + r, \quad -\phi''(r) = \frac{w+1}{r}.$$

The Gaussian width $\sigma = \sqrt{r/(w+1)}$ tends to infinity.

6.2 Uniform localization and remainders

The exact identity

$$\phi(r y) - \phi(r) = -r\{w(y - 1 - \log y) + (y \log y - y + 1)\} \quad (38)$$

controls both tails: both bracketed functions are nonnegative and vanish only at $y = 1$. It also absorbs every fixed power coming from amplitudes and their derivatives.

With $t = r + \sigma u$, the exponent becomes

$$\phi(t) - \phi(r) = -u^2/2 + \sum_{h \geq 1} \gamma_h(w) r^{-h/2} u^{h+2}, \quad (39)$$

$$\gamma_h(w) = \frac{(-1)^{h+1}((h+1)w+1)}{(h+1)(h+2)(w+1)^{(h+2)/2}}. \quad (40)$$

For each fixed h , these coefficients are bounded for $w \geq 1$, as are the normalized amplitude coefficients. On the symmetric interval $|u| \leq r^\delta$, with a sufficiently small fixed $\delta > 0$, Taylor expansion through degree $2J - 1$ in $r^{-1/2}$ has, after including the Gaussian factor, a remainder bounded by

$$C_J r^{-J} P_J(|u|) e^{-u^2/4},$$

where P_J is a fixed polynomial. Its integral is $O_J(r^{-J})$. Odd terms integrate to zero. The rest of the fixed relative neighborhood is Gaussian-small, and (38) handles the exterior. This proves a uniform algebraic expansion without residual powers of $\log r$.

To pass from integrals to sums, apply Euler–Maclaurin to

$$f_\ell(t) = t^{-\ell-2} e^{\phi(t)}.$$

For fixed m, ℓ , its endpoint derivatives vanish once n is large enough, and Gaussian rescaling and the derivative product rule give

$$\int_0^\infty |f_\ell^{(2m)}(t)| dt \leq C_{m,\ell} \sigma^{-2m} e^{\phi(r)} r^{-\ell-2} \sigma. \quad (41)$$

On the exterior, (38) absorbs the finitely many derivative powers of n and $1/t$; near zero take $n > \ell + 2 + 2m$. Hence the relative lattice error is $O_m(((w+1)/r)^m)$. Taking $m = J + 1$ makes it $o(r^{-J})$, since $w = \log r - 1$. There is no fractional-power lattice correction.

The leading Gaussian integral is

$$\frac{1}{2\pi e^2} r^{-2} e^{\phi(r)} \sqrt{\frac{2\pi r}{w+1}} = M_n.$$

The preceding bounds applied to the residual term in (35) prove (3).

6.3 An explicit algorithm for every coefficient

Let V be a centered Gaussian variable of variance $1/(w+1)$, used only as a device for exact polynomial moments. Then the coefficients in Theorem 1.1 are

$$c_j(w) = [z^{2j}] \mathbb{E} \left[\sum_{\ell \geq 0} \alpha_\ell z^{2\ell} (1 + Vz)^{-\ell-2} \times \exp \left\{ \sum_{h \geq 1} \frac{(-1)^{h+1}((h+1)w+1)}{(h+1)(h+2)} V^{h+2} z^h \right\} \right]. \quad (42)$$

For fixed j all sums truncate finitely. Use $\mathbb{E}V^{2m} = (2m-1)!!/(w+1)^m$ and $\mathbb{E}V^{2m+1} = 0$. Counting the Gaussian powers gives the denominator and degree bounds asserted in the theorem.

For example the amplitude t^{-2} and the cubic/quartic exponent terms give

$$c_1 = \frac{5}{6} + \frac{3}{w+1} - \frac{2w+1}{(w+1)^2} - \frac{3w+1}{4(w+1)^2} + \frac{5(2w+1)^2}{24(w+1)^3},$$

which simplifies to (4); the next coefficient is (5). The package gives c_3, c_4 and the exact generator.

6.4 The Bell polynomial form

Dobiński's identity reads

$$B_n(e) = e^{-e} \sum_{k \geq 0} \frac{k^n e^k}{k!}.$$

Stirling's formula and the same saddle give

$$B_n(e) \sim \frac{e^{-e} r^n \exp(r-n)}{\sqrt{w+1}}.$$

Comparing with M_n proves the second equivalent in (2), with constant

$$\frac{e^{e-2}}{\sqrt{2\pi}} = 0.818193265211777 \dots$$

This completes the proof of Theorem 1.1.

7 Canonical continuous inversion

7.1 The increasing branch

The spectral measure gives a canonical interpolation

$$\mathcal{A}(z) = \sum_{\lambda: D(\lambda)=0} \omega_\lambda \lambda^z, \quad \lambda^z = \exp(z \log \lambda). \quad (43)$$

It is entire in z : the support has a positive minimum and all positive moments, so the series is locally uniformly dominated on each vertical strip. For real x , $\mathcal{A}(x) > 0$, and

$$(\log \mathcal{A})''(x) = \text{Var}_x(\log \lambda) > 0,$$

where the variance is under the tilted probability weights. The measure is nondegenerate since $a_1 = 1$ and $a_2 = 2$. Jensen's inequalities give

$$\mathcal{A}'(0) = \mathbb{E} \log \lambda < 0, \quad \mathcal{A}'(1) = \mathbb{E}(\lambda \log \lambda) > 0.$$

Thus the unique minimum lies in $(0, 1)$; $\mathcal{A}$ is strictly increasing on $[1, \infty)$, from $\mathcal{A}(1) = 1$ to infinity. Let $x_*(L) \geq 1$ be its inverse defined by $\mathcal{A}(x_*) = e^L$ for $L \geq 0$. All preceding asymptotic arguments hold for real $x \rightarrow \infty$.

7.2 Controlled inversion of the full expansion

Write $r = r(x)$ and $w = w(x)$ as in (1), and set

$$F_0(x) = x \log r - x + r = e^{w+1}(w^2 + 1), \quad (44)$$

$$g(x) = \frac{3}{2} \log r + \frac{1}{2} \log(w + 1) + 2 + \frac{1}{2} \log(2\pi). \quad (45)$$

Let $\ell_j(w) = [z^j] \log(\sum_{k \geq 0} c_k(w) z^k)$, and define

$$H_J(x) = F_0(x) - g(x) + \sum_{j=1}^J \frac{\ell_j(w(x))}{r(x)^j} \quad (J \geq 0). \quad (46)$$

Then $\log \mathcal{A}(x) = H_J(x) + O_J(r^{-J-1})$ and

$$H'_J(x) = w + 1 + O(1/[r(w + 1)]) > 0$$

for large x .

Theorem 7.1. *If $x_J(L)$ solves $H_J(x_J) = L$ on its eventual increasing branch, then*

$$x_*(L) = x_J(L) + O_J\left(\frac{r(x_J)^{-J-1}}{w(x_J) + 1}\right) \quad (L \rightarrow \infty). \quad (47)$$

Proof. No differentiation of an asymptotic remainder is needed. Convexity bounds $(\log \mathcal{A})'(x)$ between the secants on $[x - 1, x]$ and $[x, x + 1]$. The already proved expansion on those intervals gives

$$(\log \mathcal{A})'(x) = w + 1 + O(1/r).$$

At x_J the logarithmic residual is $O_J(r^{-J-1})$. The mean-value theorem and the eventual positive lower bound on the derivative prove (47). $\square$

7.3 An explicit inverse based on the exact leading core

Let $w_0 > 0$ solve

$$e^{w_0+1}(w_0^2 + 1) = L, \quad r_0 = e^{w_0+1}, \quad N_0 = r_0 w_0. \quad (48)$$

This core equation is strictly increasing for $w_0 > 0$. Define

$$d_0 = \frac{3}{2} + \frac{\frac{1}{2} \log(w_0 + 1) + 2 + \frac{1}{2} \log(2\pi)}{w_0 + 1}. \quad (49)$$

Since $F'_0(x) = w + 1$, Taylor expansion yields

$$x_*(L) = N_0 + d_0 + O(1/[r_0(w_0 + 1)]). \quad (50)$$

A further explicit term is

$$x_*(L) = N_0 + d_0 - \frac{1}{r_0(w_0 + 1)} \left[c_1(w_0) + \frac{d_0^2}{2(w_0 + 1)} - d_0 \left\{ \frac{3}{2(w_0 + 1)} + \frac{1}{2(w_0 + 1)^2} \right\} \right] + O(1/[r_0^2(w_0 + 1)]). \quad (51)$$

Indeed $F_0'' = 1/[r(w+1)]$ and $g' = r^{-1}\{3/[2(w+1)] + 1/[2(w+1)^2]\}$; substituting $N_0 + d_0$ leaves exactly the displayed logarithmic residual of order r_0^{-1} . All coefficients and the needed derivatives remain bounded for $w_0 \geq 1$.

For numerical work, $2W(\frac{1}{2}\sqrt{L}/e)$ initializes safeguarded Newton for (48); its derivative is $e^{w+1}(w+1)^2$. Newton can then solve (46). A numerical residual η bounds the numerical error in the H_J root by order $|\eta|/(w+1)$. A certified enclosure for the *true* inverse at a finite target additionally requires an explicit numerical bound for the asymptotic remainder. The O_J estimates above prove error scales, not ready-made interval certificates.

For $Y > 1$, the discrete threshold is $\min\{n : a_n \geq Y\} = \lceil x_*(\log Y) \rceil$. If a finite-target enclosure straddles an integer, neighboring exact recurrence values must be compared. A finite expansion purely in inverse logarithms is not claimed to give integer accuracy uniformly.

8 The exponential generating function on the positive axis

Let $\mathcal{E}(z) = \sum_{n \geq 0} a_n z^n / n!$, distinguished from the interpolation $\mathcal{A}$ and the normalized Bessel numerator E .

Corollary 8.1. *The function $\mathcal{E}$ is entire and has the exact representation*

$$\mathcal{E}(z) = \sum_{\lambda} \omega_{\lambda} \exp(\lambda z).$$

For real $z \rightarrow +\infty$, put $q = \exp(z+1)$. Then

$$\mathcal{E}(z) = \frac{e^{-2}}{\sqrt{2\pi}} e^q q^{-3/2} \left(1 + \frac{67}{24q} + \frac{11857}{1152q^2} + O(q^{-3}) \right). \quad (52)$$

There is an expansion to every fixed inverse power of q ; its coefficients are $c_j(0)$ from (42).

Proof. The weight estimate (27) makes the exponential spectral sum locally uniformly convergent for complex z . Its Taylor coefficients are the exact moments, proving entirety and the representation. For $k \geq \sqrt{q}$, the pole displacement in $\exp(z\lambda_k)$ is smaller than every algebraic relative error. The complementary spectrum contributes at most $\exp(2z\sqrt{q})$ for large q , negligible compared with $e^q q^{-3/2}$.

Using (36), the main sum is

$$\frac{e^{-2}}{\sqrt{2\pi}} e^q \mathbb{E} \left[K^{-3/2} \left(1 + \frac{11}{12K} + \frac{313}{288K^2} + \cdots \right) \mathbf{1}_{\{K \geq 1\}} \right],$$

where K is Poisson with mean q . Poisson concentration and Taylor expansion about q justify this to every fixed order. For general fixed $p > 0$,

$$\mathbb{E}[K^{-p} \mathbf{1}_{\{K \geq 1\}}] = q^{-p} \left(1 + \frac{p(p+1)}{2q} + \frac{p(p+1)(p+2)(3p+5)}{24q^2} + O(q^{-3}) \right).$$

The first correction is $15/8 + 11/12 = 67/24$; the next is $665/128 + (11/12)(35/8) + 313/288 = 11857/1152$. Alternatively the real saddle of $(z+2)t - t \log t$ is q . Retain the lower cutoff $t \geq \sqrt{q}$ (or a smooth equivalent): the uncut inverse-power amplitude is not integrable at zero when $w = 0$. All cutoff endpoint terms are $\exp(O(\sqrt{q} \log q))$, negligible relative to e^q . The central Gaussian coefficient and Taylor bounds remain bounded at $w = 0$, and give exactly $c_j(0)$ from (42). Euler–Maclaurin on the cut-off domain has negligible endpoints and proves the stated all-orders form. $\square$

9 Numerical checks and reproducibility

The exact recurrence used for validation is

$$p_{0,0} = 1, \quad p_{n,0} = 0 \ (n > 0), \quad p_{n,k} = p_{n-1,k-1} + k \sum_{j \geq k} p_{n-1,j}, \quad a_n = \sum_k p_{n,k}. \quad (53)$$

Suffix sums evaluate a row in linear time, using exact integers. Let $R_n = a_n/M_n$. The following high-precision diagnostics use exact a_n through $n = 2500$:

n	R_n	$r(R_n - 1)$	$c_1(w)$
50	1.058106438359	1.352791674642	1.251473654579
100	1.032686382461	1.240030857333	1.186048117956
200	1.018353693325	1.165057005282	1.135943310080
400	1.010254878707	1.112627221355	1.096857105828
800	1.005694089255	1.074346527794	1.065802240138
1600	1.003140908882	1.045321581963	1.040701211579
2500	1.002134095265	1.030095275457	1.026990044234

The second scaled residual $r^2(R_n - 1 - c_1/r)$ likewise approaches $c_2(w)$; its discrepancy decreases from about 0.244 at $n = 50$ to 0.00638 at $n = 2500$.

A separate spectral computation with 110 decimal digits and 65 nodes begins with

	node λ	probability weight ω
0	0.320955976594004859	0.541638005518968697
1	1.115304216978757830	0.256514749784783314
2	2.042059059595086477	0.118696820250535227
3	3.014766264330034697	0.051578277597176440

The sum of these 65 weights agrees with 1 at the displayed 50 digits in the accompanying output. Their moments agree with the exact recurrence with relative errors from about 1.46×10^{-65} at $n = 1$ to 1.84×10^{-40} at $n = 30$, consistent with the omitted positive tail. This root search is a numerical diagnostic; the analytic proof, rather than the scan, establishes completeness of the spectrum.

The inverse check solves $H_J(x) = \log a_n$ for $J = 0, 1, 2$ and exact targets through $n = 500$. At $n = 500$, the $J = 2$ approximation differs from 500 by 2.07456×10^{-7} . Its error multiplied by $r^3(w + 1)$ is 2.18743, compared with the next logarithmic coefficient 2.13558.

The reproducibility archive contains the editable L^AT_EX source, a portable build script, exact coefficient generators, all three numerical tests, their outputs, dependency versions, a single replay command, and mathematical verification records. It contains no third-party paper copies. Numerical arithmetic uses `mpmath`; symbolic coefficients use `SymPy`. The formulas are derived symbolically, never fitted to these data.

[write, 2 October 2026] In this repository the five programs and their outputs are shipped under `code/57-catalan-*` and `data/57-catalan-*`, with the verification and validation records as `57-catalan-*.md`; the delivered PDF and checksum manifest are not shipped, and the delivered L^AT_EX source is the text of this Part. The README lists every file and gives a rerun recipe that leaves the shipped outputs untouched.

10 Literature scope and further questions

Elizalde’s Proposition 6.1 and Corollary 6.2 [2] give coarse bounds and subfactorial growth; Section 7 leaves precise asymptotics for 1-23-4 open. Callan [3], Duchi–Guerrini–Rinaldi [5], and Beaton et al. [4] supply the models used here. Later work by Lin–Fu [6], Cervetti [7], Frosini–Guerrini–Rinaldi [8], and Testart [9] provides refinements, generating trees, or related bijections. The sources surveyed do not supply a previous precise equivalent of the form (2). As of the dated review, the OEIS entry lists the known fraction but no sharp asymptotic. These observations are a scoped literature statement, not a proof of global novelty.

[write, 2 October 2026] In this repository, the neighbouring report `a113226-vincular-avoiders` (in the same collection directory) proves precise asymptotics for the vincular pattern 12-34 (OEIS A113226), a different sequence, by a closed exponential generating function; it also cites Elizalde’s bounds and Callan’s 1-23-4 bijection paper. The two reports share no theorem or method and are kept separate; see the Guide, “Relation to other reports”. The Bell-number analogue of the saddle used here is treated in `a277364-bell-asymptotics` and in the repository’s transseries volume (Guide, “Relation to other reports”).

The formal fraction, general continued-fraction techniques, Bessel identities, and saddle methods are established tools. The central analytic issue addressed by this report is their connection to the *correct global positive measure*; a formal quotient alone would not justify the moment asymptotics. Several further questions remain meaningful.

1. Can the exact Bessel nodes and weights be enclosed by efficient certified interval algorithms with explicit tail bounds? This would turn the asymptotic inverse estimates into finite-target enclosures.
2. What is the optimally truncated remainder of the algebraic expansion, and how do exponentially small node displacements interact with lattice effects? The present result is an all-orders Poincaré expansion, not an exponentially complete transseries.
3. Can the positive spectral representation be deformed to refined powered-Catalan statistics, with a uniform two-parameter asymptotic theorem?
4. Is there a direct probabilistic or combinatorial explanation for the Bell-polynomial parameter e and the factor $r^{-3/2}$?

[write, 2 October 2026] Question 2 is now partly answered. Part II treats the lattice effects at every fixed exponential order: after the node displacements are removed with relative error $O(e^{-dn})$, $d < \log 2$ (Proposition 12.1), Poisson summation turns the lattice sum into exact Fourier sectors, and any fixed number of them, each with its own all-orders expansion, approximates a_n up to the next envelope (Theorem 11.1). Still open: the optimally truncated remainder, any statement with a growing number of sectors, a Borel sum, and an exponentially complete transseries, all of which Part II disclaims (Section 17). Questions 1, 3 and 4 are untouched; Part II’s numerical checks are not certified either.

Part II

Fixed Fourier sectors for the powered Catalan numbers

Source. Batch-77 manuscript 56, archive `oeis-powered-catalan-fourier-addendum.zip`, main file `powered-catalan-fourier-addendum.tex` (497 lines, 10-page PDF), dated 1 October 2026. Author line: “A research addendum with reproducible diagnostics”. No pin. An addendum to source 57 (Part I), printed in full, with labels prefixed `pcn:fs;`; its citations of source 57 are now references to Part I, and three symbols are renamed (Guide, “Notation”): $\mathcal{B}_m$ for its β_m , β_{2q} for its Bernoulli numbers B_{2q} , and E_S for its Schläfli remainder E .

Abstract (as delivered). The positive Bessel spectrum of the powered Catalan numbers permits a fixed-mode Fourier refinement beyond their all-orders algebraic expansion. We prove a zero-free complex wedge for the exact Bessel weight, replace the full spectral sum by a lattice sum with relative error $O(e^{-dn})$ for every $d < \log 2$, and control the entire omitted Fourier tail by the next complex-saddle envelope. Each retained exact sector has its own all-orders expansion with the same rational coefficients as the real saddle. Exact sectors and finite asymptotic truncations are kept distinct: a fixed algebraic approximation to the principal sector does not have exponentially improved accuracy.

11 Definitions and the fixed sector theorem

Let a_n be OEIS A113227, beginning $1, 1, 2, 6, 23, 105, 549, \dots$. We use the positive discrete spectral representation established in Part I (Theorem 3.1 and Proposition 4.2). Set

$$w = W_0(n/e), \quad r = n/w = e^{w+1},$$

and, with principal logarithms and square roots in the right half-plane, define

$$b(z) = \sqrt{z} Y_{z+1}(2\sqrt{z}) - Y_z(2\sqrt{z}), \quad Q(z) = \frac{1}{\pi^2 z b(z)^2}.$$

The letter Q distinguishes the exact weight from Lambert W . Use the explicit half-integer endpoints

$$a = \lfloor r/8 \rfloor + \frac{1}{2}, \quad B = \lfloor 4r \rfloor + \frac{1}{2},$$

and the exact finite-window Fourier integrals

$$I_j(n) = \int_a^B Q(t) t^n e^{2\pi i j t} dt, \quad j \in \mathbb{Z}. \quad (54)$$

For every fixed $j \geq 0$, let

$$w_j = W_{-j}(n/e), \quad t_j = \frac{n}{w_j}, \quad M_j = \frac{e^{-2t_j^{n-3/2}} \exp(t_j - n)}{\sqrt{2\pi(w_j + 1)}}, \quad E_j = |M_j|. \quad (55)$$

The square root is the continuation of the positive root, equivalently the principal one for sufficiently large n . In particular $M_0 = E_0$ is the real leading equivalent proved in Part I: it is M_n of (1), and Theorem 1.1 gives $a_n \sim M_n$.

Theorem 11.1. *For every fixed integer $J \geq 0$ and every $0 < d < \log 2$,*

$$a_n = I_0(n) + 2\Re \sum_{j=1}^J I_j(n) + O_J(E_{J+1}) + O_{J,d}(M_0 e^{-dn}). \quad (56)$$

For every fixed $j \geq 0$ and $L \geq 1$,

$$I_j(n) = M_j \left\{ \sum_{\ell=0}^{L-1} \frac{c_\ell(w_j)}{t_j^\ell} + O_{j,L}(r^{-L}) \right\}, \quad (57)$$

where the rational functions c_ℓ are exactly those of the real saddle in Part I (Theorem 1.1 and the algorithm (42)). The first two are

$$c_0(u) = 1, \quad c_1(u) = \frac{20u^3 + 86u^2 + 128u + 67}{24(u+1)^3}.$$

For every fixed $j \geq 1$,

$$\log \frac{E_j}{M_0} = -\frac{2\pi^2 j^2 r}{w+1} + O_j(r/w^3) = -\frac{2\pi^2 j^2 n}{w^2} (1 + O_j(w^{-1})). \quad (58)$$

Thus $E_{j+1} = o(E_j)$, and $M_0 e^{-dn} = o(E_j)$ for every fixed $d > 0$.

The proof separates three errors. Spectral replacement costs $O_d(M_0 e^{-dn})$ for all $d < \log 2$. The real-window and paired vertical-edge errors are $O(M_0 e^{-\gamma n})$ with a fixed $\gamma > \log 2$, for example $\gamma = 1$. The remaining horizontal Fourier sum is $O_J(E_{J+1})$. Constants can depend on the fixed mode cutoff and truncation order.

Remark 11.2. The real pair $2\Re I_j$ is oscillatory and can vanish; its error is measured against the positive envelope E_j , not relative to the real pair. Formula (56) involves the exact Bessel integrals (54). A fixed algebraic truncation of I_0 has error $O(M_0 r^{-L})$, larger than every fixed nonzero Fourier envelope. Therefore substituting finitely many terms into all retained sectors does not by itself give an exponentially accurate numerical approximation.

The statements also hold for real $n \rightarrow \infty$ if a_n means the canonical positive spectral interpolation (43) of Part I. This extension is a statement about counts; no exponentially accurate inverse is claimed here.

12 Global exponentially accurate spectral replacement

The audited spectrum of Part I gives (Proposition 4.2, equations (24), (26), (27) and (15)), for every fixed $0 < c < 1$ and all sufficiently large integers k ,

$$|\lambda_k - k| \leq e^{-ck \log k}, \quad \left| \frac{\omega_k}{Q(k)} - 1 \right| \leq C_c e^{-ck \log k}, \quad (59)$$

$$0 < Q(k) \leq C k^{-2} \exp(-k \log k + 2k), \quad a_n = \sum_{\lambda} \omega_{\lambda} \lambda^n. \quad (60)$$

All weights are positive, their sum is one, and the finitely many excluded nodes lie in a bounded interval. Fixed multiplicative constants in the first estimate may be absorbed by decreasing c .

Proposition 12.1. *For a sufficiently large fixed integer K , and every $0 < d < \log 2$,*

$$a_n = \sum_{k \geq K} Q(k)k^n + O_d(M_0 e^{-dn}).$$

Proof. Split at $k = \sqrt{n}$. By positivity and total mass one, all lower spectral nodes together contribute at most $(2\sqrt{n})^n$ for large n . The corresponding lower lattice sum has the same bound up to a fixed constant, because $\sum_{k \geq K} Q(k) < \infty$. Since

$$\log M_0 = n \log n - n \log \log n - n + O(n \log \log n / \log n),$$

these terms are $M_0 \exp[-(1/2 + o(1))n \log n]$.

For $k \geq \sqrt{n}$, $n|\lambda_k - k|/k \rightarrow 0$ uniformly. The mean-value estimate for $(\lambda_k/k)^n$, together with (59), gives

$$|\omega_k \lambda_k^n - Q(k)k^n| \leq C_c(n+1)Q(k)k^n e^{-ck \log k}.$$

Put $p = 1 + c$ and $F_p(x) = n \log x - px \log x + 2x$. This is strictly concave and has its unique maximum at

$$\frac{n}{s} = p \log s + p - 2, \quad s = \frac{n}{p W_0((n/p) e^{1-2/p})}.$$

Consequently $s/r = p^{-1}(1 + O_p(w^{-1}))$ and

$$F_p(s) = n \log s - n + ps, \quad F_p(s) - F_1(r) = -n \log p + O_p(n/w). \quad (61)$$

Up to n^2 there are at most n^2 summands, each bounded by $\exp F_p(s)$. For $k > n^2$, eventually $F_p(k) \leq -(p/2)k \log k$, so that tail is smaller than one. Polynomial factors and the prefactor between $\exp F_1(r)$ and M_0 are harmless. Thus the full error is

$$M_0 O_c(\exp[-n \log(1+c) + O_c(n/w)]).$$

For a given $d < \log 2$, choose $c < 1$ such that $\log(1+c) > d$. This proves the claim globally, rather than only near the principal saddle. $\square$

13 The complex Bessel weight

Lemma 13.1. *For every fixed $0 < \theta < \pi/2$ and $N \geq 1$, uniformly as $|z| \rightarrow \infty$ in $|\arg z| \leq \theta$,*

$$b(z) = -\frac{e}{\pi} z \Gamma(z) z^{-z/2} \left(\sum_{m=0}^{N-1} \mathcal{B}_m z^{-m} + O_{N,\theta}(|z|^{-N}) \right), \quad (62)$$

where $\mathcal{B}_0 = 1$. In particular b is eventually zero-free there, and

$$Q(z) = \frac{z^{-2}}{2\pi e^2} e^{-z \operatorname{Log} z + 2z} \left(\sum_{m=0}^{N-1} \alpha_m z^{-m} + O_{N,\theta}(|z|^{-N}) \right). \quad (63)$$

The coefficients agree with the real expansions, including

$$\mathcal{B}_1 = -\frac{1}{2}, \quad \mathcal{B}_2 = -\frac{5}{24}, \quad \alpha_0 = 1, \quad \alpha_1 = \frac{5}{6}, \quad \alpha_2 = \frac{73}{72}.$$

Each order- N normalized remainder is holomorphic; on a smaller closed wedge its q th derivative is $O_{N,q,\theta}(|z|^{-N-q})$.

13.1 Schl\"afli representation and contour rotation

Write $z = Re^{i\phi} = u + iv$, so $u \geq R \cos \theta$. Schl\"afli's exact formula [11, 10.9.7] is

$$Y_\nu(\zeta) = \frac{1}{\pi} \int_0^\pi \sin(\zeta \sin s - \nu s) ds - \frac{1}{\pi} \int_0^\infty (e^{\nu s} + \cos(\pi\nu)e^{-\nu s}) e^{-\zeta \sinh s} ds, \quad |\arg \zeta| < \pi/2.$$

All parameter integrals here converge locally uniformly in $\Re z > 0$. Combining the growing terms and putting $\xi = \sqrt{z} e^s$ gives

$$b(z) = -\frac{z^{-z/2}}{\pi} \int_{\sqrt{z}}^{\infty e^{i\phi/2}} (\xi - 1) \xi^{z-1} e^{-\xi + z/\xi} d\xi + E_S(z), \quad (64)$$

where

$$|E_S(z)| \leq C_\theta (1 + \sqrt{R}) e^{\pi|v| + 2\sqrt{R}}.$$

Indeed the trigonometric integral has this bound, and the decaying integrals are bounded by $1/u$ or $1/(u+1)$ before their explicit factors. Uniform Stirling gives, for $G(z) = z\Gamma(z)z^{-z/2}$,

$$|G(z)| \asymp_\theta R^{1/2} \exp\left(\frac{1}{2}u \log R - \frac{1}{2}v\phi - u\right). \quad (65)$$

Thus $E_S/G = O(\exp[-\eta R \log R])$ for any fixed $\eta < \frac{1}{2} \cos \theta$.

Rotate the integration ray in (64) from argument $\phi/2$ to argument ϕ , retaining starting radius $\sqrt{R}$. The large closing arc tends to zero since $\Re \xi \geq |\xi| \cos \theta > 0$. No cut or singularity is crossed. On the small connecting arc $\xi = \sqrt{R} e^{i\psi}$,

$$|z^{-z/2} \xi^{z-1}| = R^{-1/2} e^{v(\phi/2 - \psi)} \leq R^{-1/2}.$$

Including $\xi - 1$, the arc length, and $|e^{-\xi + z/\xi}| \leq e^{2\sqrt{R}}$ bounds the arc by $C_\theta \sqrt{R} e^{2\sqrt{R}}$ after the $z^{-z/2}$ normalization. It is again superexponentially small relative to (65).

The rotated integral, denoted $\mathcal{I}(z)$, becomes

$$\mathcal{I}(z) = z^z \int_{R^{-1/2}}^\infty (zx - 1) x^{z-1} e^{-zx + 1/x} dx. \quad (66)$$

This rotation is important: on the positive x ray, $|x^z e^{-zx}| = \exp[u(\log x - x)]$, so the absolute-value saddle is exactly at $x = 1$.

13.2 Uniform Gamma normalization and tails

Let $h(x) = x - 1 - \log x$. Stirling also gives

$$|z\Gamma(z)| \asymp_\theta R^{1/2} |z^z| e^{-u}.$$

Since $h(x) \geq c(x-1)^2$ on $[1/2, 2]$,

$$\frac{|z^z|}{|z\Gamma(z)|} \int_{1/2}^2 |zx - 1| x^{u-1} e^{-ux} dx \leq C_\theta. \quad (67)$$

On $R^{-1/2} \leq x \leq 1/2$, use $e^{1/x} \leq e^{\sqrt{R}}$ and

$$\int_0^{1/2} x^{u-1} e^{-ux} dx \leq \frac{2}{u} e^{-u} e^{-uh(1/2)}.$$

This follows by the substitution h , with $dx/x = -dh/(1-x)$ and $(1-x)^{-1} \leq 2$. After normalization, the lower tail is $O_\theta(e^{-c_\theta R})$. For $x \geq 2$, use $e^{1/x} \leq e^{1/2}$, $|zx - 1|/x \leq R + 1$, and $h'(x) \geq 1/2$. The upper tail has the same exponential bound. Hence (66), divided by $z\Gamma(z)$, can be restricted to $[1/2, 2]$ with exponential error.

13.3 Finite Gamma sums and a summable product remainder

Take $H = \lfloor R^{1/4} \rfloor$. On $[1/2, 2]$,

$$e^{1/x} = \sum_{j=0}^H \frac{x^{-j}}{j!} + O\left(\frac{e^2 2^{H+1}}{(H+1)!}\right).$$

By (67), the normalized integral error is smaller than every fixed power of R^{-1} . Extend each retained polynomial term to $(0, \infty)$; do not extend the singular untruncated $e^{1/x}$. For $j \leq H < u/4$,

$$\int_0^{1/2} x^{u-j-1} e^{-ux} dx \leq \frac{2^j}{u/2 - j} e^{-u} e^{-uh(1/2)}.$$

To verify this, differentiate $x^{u-j} e^{-ux}$ and use $u - j - ux \geq u/2 - j$. Summing against $1/j!$ costs at most e^2 . On the upper tail $x^{-j} \leq 1$, so summation costs at most e . All extension errors are exponentially small.

The complex Laplace–Gamma identity now gives

$$z^z \int_0^\infty (zx - 1) x^{z-j-1} e^{-zx} dx = z^j \{\Gamma(z+1-j) - \Gamma(z-j)\}.$$

It is valid since $\Re(z-j) > 0$ and $\Re z > 0$, with the compatible principal powers. Thus, for every fixed M ,

$$\frac{\mathcal{I}(z)}{z\Gamma(z)} = \sum_{j=0}^H \frac{1}{j!} \left\{ \prod_{\ell=0}^{j-1} (1 - \ell/z)^{-1} - \frac{1}{z} \prod_{\ell=1}^j (1 - \ell/z)^{-1} \right\} + O_{M,\theta}(R^{-M}). \quad (68)$$

Empty products equal one. For $\delta = 0, 1$, put

$$P_{j,\delta}(s) = \prod_{\ell=\delta}^{j-1+\delta} (1 - \ell s)^{-1} = \sum_{m \geq 0} p_{m,\delta}(j) s^m.$$

For fixed m , $p_{m,\delta}(j)$ is a polynomial in j of degree at most $2m$, by expanding the logarithm and using power sums. On $|s| = \rho_j = [4(j+1)^2]^{-1}$, $|P_{j,\delta}(s)| \leq e$. For $j \leq H$, eventually $R^{-1} \leq \rho_j/2$. Cauchy's coefficient bound and a geometric sum give

$$\left| P_{j,\delta}(1/z) - \sum_{m=0}^{N-1} p_{m,\delta}(j) z^{-m} \right| \leq 2e 4^N (j+1)^{2N} R^{-N}. \quad (69)$$

This is summable against $1/j!$, uniformly in R . Extending each coefficient sum from $j \leq H$ to all j costs another factorial tail. We obtain

$$\frac{\mathcal{I}(z)}{z\Gamma(z)} = e \sum_{m=0}^{N-1} \mathcal{B}_m z^{-m} + O_{N,\theta}(R^{-N}),$$

where the coefficientwise generator is

$$\mathcal{B}(s) = e^{-1} \sum_{j \geq 0} \frac{1}{j!} \left\{ \prod_{\ell=0}^{j-1} (1 - \ell s)^{-1} - s \prod_{\ell=1}^j (1 - \ell s)^{-1} \right\} = \sum_{m \geq 0} \mathcal{B}_m s^m. \quad (70)$$

Equation (70) means coefficientwise summation, not convergence as an analytic function of s .

13.4 Zero-freeness and analytic remainder bounds

Combining the preceding estimates proves (62). With $N = 1$, its normalized factor is $1 + O(1/R)$, so its modulus is eventually at least $1/2$. The remaining factors do not vanish. This proves eventual zero-freeness. Algebraically,

$$Q(z) = e^{-2} \frac{z^{z-3}}{\Gamma(z)^2} \left(\sum_{m=0}^{N-1} \mathcal{B}_m z^{-m} + O(R^{-N}) \right)^{-2}.$$

Uniform Stirling [11, §5.11] proves (63), with formal coefficient generator

$$\sum_{m \geq 0} \alpha_m s^m = \mathcal{B}(s)^{-2} \exp \left[- \sum_{q \geq 1} \frac{\beta_{2q} s^{2q-1}}{q(2q-1)} \right]. \quad (71)$$

For derivative estimates, choose a larger wedge with angle still below $\pi/2$. A disk of radius $\epsilon|z|$ centered in the smaller wedge lies in the larger one, with all moduli comparable to $|z|$. Cauchy's formula gives the asserted derivative bounds for the analytic normalized remainder. The nonanalytic cutoff H was used only to establish the bound and is never differentiated. In particular,

$$\frac{Q'(z)}{Q(z)} = 1 - \text{Log } z - \frac{2}{z} + O(|z|^{-2}). \quad (72)$$

This completes Lemma 13.1.

14 Poisson summation and the entire omitted Fourier tail

14.1 The real window

Let $\phi(x) = n \log x - x \log x + 2x$. For fixed $c > 0$,

$$\phi(cr) - \phi(r) = -n\mathcal{H}(c) + r\{c(1 - \log c) - 1\}, \quad \mathcal{H}(c) = c - 1 - \log c. \quad (73)$$

Here $\mathcal{H}(1/8) = \log 8 - 7/8 > 1.20$ and $\mathcal{H}(4) = 3 - \log 4 > 1.61$. Outside the window, ϕ' has the appropriate sign and magnitude at least a positive constant times w . Concavity bounds each discrete tail by a geometric sum from its boundary. Polynomial amplitudes and $O(1)$ endpoint rounding cost only $\exp(o(n))$. Therefore, for any fixed $\gamma < \min(\mathcal{H}(1/8), \mathcal{H}(4))$,

$$\sum_{k \geq K} Q(k)k^n = \sum_{a < k < B} Q(k)k^n + O(M_0 e^{-\gamma n}). \quad (74)$$

We may and will choose $\gamma = 1$.

Put $f(z) = Q(z)z^n$. Finite-interval Poisson summation gives

$$\sum_{a < k < B} f(k) = I_0 + 2 \sum_{j \geq 1} \Re I_j. \quad (75)$$

The paired series converges absolutely. Indeed, the first integration by parts produces the boundary term

$$\frac{(-1)^j}{2\pi i j} \{f(B) - f(a)\},$$

which is purely imaginary. A second integration by parts bounds the remaining real part by C_n/j^2 . Alternatively, ordinary Fourier convergence of the piecewise smooth periodicization at zero proves (75); no jump occurs there since both endpoints are half-integers.

14.2 One common contour height

Fix $m = J + 1 \geq 1$, and write

$$\psi_j(z) = n \operatorname{Log} z - z \operatorname{Log} z + 2z + 2\pi i j z.$$

The saddle equation is

$$n/t_j = \operatorname{Log} t_j - 1 - 2\pi i j = w_j,$$

with branch specified by $w_j + \operatorname{Log} w_j = \log(n/e) - 2\pi i j$. For fixed j ,

$$w_j = w - 2\pi i j + O_j(w^{-1}), \quad t_j/r = 1 + O_j(w^{-1}).$$

For $j \geq 1$, $\Im t_j > 0$. In particular

$$h = \Im t_m \sim \frac{2\pi m r}{w + 1} \longrightarrow \infty.$$

Every rectangle with real part between a and B and height h lies eventually in any fixed positive-angle wedge. Lemma 13.1 therefore permits shifting every mode $j \geq m$ to this same height. Cauchy's theorem, with its orientation retained, gives

$$\begin{aligned} I_j &= H_j + i(-1)^j \int_0^h e^{-2\pi j y} \{f(a + iy) - f(B + iy)\} dy, \\ H_j &= \int_a^B f(x + ih) e^{2\pi i j(x + ih)} dx. \end{aligned} \tag{76}$$

14.3 Paired vertical endpoints

Since $f(a)$ and $f(B)$ are real, for either endpoint x ,

$$|\Im f(x + iy)| \leq y \sup_{0 \leq v \leq h} |f'(x + iv)|.$$

For $0 \leq v \leq h$ and x a fixed positive multiple of r ,

$$\Re \phi(x + iv) - \phi(x) = O_J(n/w^2).$$

This follows directly from $v/x = O_J(w^{-1})$ in the analytic phase. Equations (63) and (72) give $|f'| \leq C_J w |f|$ on the edges. Combining with (73), and absorbing subexponential factors, gives

$$\sup_{0 \leq v \leq h} |f'(x + iv)| \leq M_0 e^{-\gamma n}$$

for every fixed $\gamma < \min(\mathcal{H}(1/8), \mathcal{H}(4))$ and sufficiently large n . The sum of the absolute real parts of the vertical pieces in (76) is finite, because

$$\int_0^h \frac{y e^{-2\pi m y}}{1 - e^{-2\pi y}} dy \leq \int_0^\infty \frac{y e^{-2\pi m y}}{1 - e^{-2\pi y}} dy < \infty. \tag{77}$$

Thus the complete paired vertical tail is $O_J(M_0 e^{-\gamma n})$. The factor y from real pairing removes the $1/y$ singularity at zero. No absolutely convergent unsigned series of the original complex I_j is asserted.

14.4 A global horizontal Gaussian bound

The horizontal terms do admit an absolute geometric bound:

$$\sum_{j \geq m} |H_j| \leq \frac{1}{1 - e^{-2\pi h}} \int_a^B |f(x + ih)| e^{-2\pi m h} dx. \quad (78)$$

At $x_m = \Re t_m$, the derivative of $\Re \psi_m(x + ih)$ vanishes. Throughout the full window,

$$\Re \psi_m''(x + ih) = -\frac{n(x^2 - h^2)}{(x^2 + h^2)^2} - \frac{x}{x^2 + h^2} \leq -c_J w/r \quad (79)$$

for all sufficiently large n , since $h/a \rightarrow 0$. The saddle is therefore the unique global maximum on this segment. Since $|x + ih| \asymp r$, Lemma 13.1 and the Gaussian bound yield

$$\int_a^B |f(x + ih)| e^{-2\pi m h} dx \leq C_J r^{-2} e^{\Re \psi_m(t_m)} \sqrt{r/w} \leq C_J E_m.$$

The prefactor in (78) tends to one. Proposition 12.1, (74), and (75)–(79) prove (56). This controls infinitely many omitted modes without summing infinitely many saddle expansions.

15 The all orders expansion of each exact sector

For fixed $j \geq 1$, shift only I_j to height $\Im t_j$. Its two vertical pieces separately are $O(M_0 e^{-\gamma n})$; the preceding endpoint bound suffices without pairing. For $j = 0$, stay on the real axis. On either horizontal segment, the same global Gaussian localization applies.

Near the saddle, write $z = t_j + u$ with u real. Then

$$A_j = -\psi_j''(t_j) = \frac{w_j + 1}{t_j}, \quad \Re A_j \asymp w/r > 0,$$

and for $q \geq 3$,

$$\psi_j^{(q)}(t_j) = (-1)^{q-1} (q-2)! \frac{(q-1)w_j + 1}{t_j^{q-1}}. \quad (80)$$

Here is a uniform fixed-order remainder argument. Put $\sigma = \sqrt{r/w}$, $\rho = (rw)^{-1/2}$, and $u = \sigma v$. The quadratic coefficient $A_j \sigma^2$ stays in a compact subset of $\{\Re z > 0\}$. The coefficient of v^{h+2} in the nonquadratic exponent is $O_h(\rho^h)$, and $u/t_j = O(\rho v)$.

On $|v| \leq \rho^{-\epsilon}$, for sufficiently small fixed $\epsilon > 0$, the nonquadratic exponent has size at most $C\rho|v|^3 = o(1)$. Expand the phase to sufficiently high degree and expand its exponential by total powers of ρ . Retaining weights below $2N$ has remainder bounded by

$$C_N \rho^{2N} P_N(|v|) \exp(C\rho|v|^3),$$

for a fixed polynomial P_N . The phase Taylor remainder has the same type of bound by analyticity on $|u| \leq \eta r$ and (80). These bounds are integrable against e^{-cv^2} . Outside the central interval, global Gaussian localization gives $\exp(-c\rho^{-2\epsilon})$, smaller than every fixed power of r^{-1} even after polynomial factors.

Apply this to the amplitude term $\alpha_\ell t_j^{-\ell} (1 + u/t_j)^{-\ell-2}$ with $N = L - \ell$. Its error is

$$O(r^{-\ell} \rho^{2(L-\ell)}) = O(r^{-L} w^{-(L-\ell)}) = O(r^{-L}).$$

The remainder of (63) itself contributes $O(r^{-L})$ by the absolute Gaussian bound. Odd total weights have odd degree in v and vanish on symmetric extension to the full real line; that extension has superalgebraically small error. Summing over the finitely many $\ell < L$ establishes the relative remainder in (57).

For $\Re A > 0$, analytic continuation of real Gaussian moments gives

$$\int_{\mathbb{R}} u^{2q} e^{-Au^2/2} du = \sqrt{2\pi} (2q-1)!! A^{-q-1/2},$$

with zero odd moments. Substitution reproduces precisely the real coefficient algorithm evaluated at w_j . The leading factor is

$$\frac{t_j^{-2}}{2\pi e^2} e^{\psi_j(t_j)} \sqrt{\frac{2\pi t_j}{w_j + 1}} = M_j,$$

since $\psi_j(t_j) = n \log t_j - n + t_j$. For each fixed j, L , the endpoint errors are smaller than $E_j r^{-L}$ and are absorbed into the same remainder.

For completeness, use formal Gaussian expectation with $\mathbb{E}V^{2q} = (2q-1)!!/(u+1)^q$ and $\mathbb{E}V^{2q+1} = 0$. Then

$$c_\ell(u) = [s^{2\ell}] \mathbb{E} \left[\sum_{k \geq 0} \alpha_k s^{2k} (1 + Vs)^{-k-2} \times \exp \left\{ \sum_{h \geq 1} \frac{(-1)^{h+1} ((h+1)u + 1) V^{h+2} s^h}{(h+2)(h+1)} \right\} \right]. \quad (81)$$

Only finitely many terms enter each coefficient. In particular,

$$c_2(u) = \frac{1168u^6 + 9968u^5 + 34692u^4 + 64536u^3 + 69380u^2 + 42208u + 11857}{1152(u+1)^6}.$$

The supplied symbolic generator computes these coefficients; there is no numerical fitting.

16 Ordering the envelopes

For bounded real v , let $t(v)$ solve

$$n/t = \log t - 1 - iv$$

by continuation from $t(0) = r$. Set $S(v) = n \log t(v) - n + t(v)$, the stationary value of the phase including ivt . Differentiation gives

$$S'(v) = it(v), \quad t'(v) = \frac{it(v)}{n/t(v) + 1}, \quad S''(0) = -\frac{r}{w+1}.$$

Repeated differentiation of the saddle relation yields $S^{(q)}(v) = O_q(r/w^{q-1})$ for $q \geq 2$, uniformly on bounded v intervals. Conjugation makes $\Re S$ even. Taylor's theorem through degree three therefore gives

$$\Re\{S(2\pi j) - S(0)\} = -\frac{2\pi^2 j^2 r}{w+1} + O_j(r/w^3).$$

The logarithm of the absolute prefactor ratio in (55) is $O_j(w^{-2})$, absorbed into the displayed remainder. This proves (58), completes Theorem 11.1, and shows that the spectral replacement error lies below every fixed Fourier sector.

17 Diagnostics and scope

The reproducibility package contains two numerical checks. The first evaluates the exact Bessel weight on central locally steepest saddle segments and compares the result with the first three rational corrections. With 160-node Gauss–Legendre quadrature at 70 decimal digits, the difference between the third scaled remainder and $c_3(w_j)$ is approximately

n	j	Scaled remainder discrepancy
10^4	0	4.89×10^{-3}
10^4	1	5.30×10^{-3}
3×10^4	0	1.76×10^{-3}
3×10^4	1	1.90×10^{-3}
10^5	0	5.76×10^{-4}
10^5	1	6.17×10^{-4}
10^8	2	9.98×10^{-7}

These non-certified diagnostics integrate central saddle segments; they do not evaluate complete cutoff-defined sectors. The proof controls the omitted contour pieces asymptotically. Separate checks recover $\alpha_3 = 11821/6480$ along arguments 0, 0.4, and 1.2 radians, and compare exact recurrence counts with a truncated exact-Bessel lattice sum.

Poisson summation, complex Gaussian expansion, and Lambert- W saddle descriptions in Bell/Touchard asymptotics are standard methods; see Paris [12] for complex Touchard saddle expansions. The work here verifies the necessary estimates for the exact powered-Catalan Bessel weight, including its zero-free wedge and the entire omitted Fourier tail. No comprehensive claim of novelty or priority is made.

[write, 2 October 2026] The repository’s transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, treats the Bell numbers ($\theta = 1$) by the same Lambert saddle (theorem `q2:thm:bell`) and, in remark `q2:rem:bell-sectors`, keeps their nonprincipal complex saddles *formal*, because identifying the Stokes multipliers of the Cauchy contour is a separate global problem. This Part proves fixed complex sectors rigorously for a different sequence, A113227, starting from an exact positive spectrum and a lattice sum rather than from a Cauchy contour. It does not settle the Bell-number question of that remark, and nothing here is claimed as new about the method.

The result fixes both the number of sectors and the algebraic order. It neither sums infinitely many asymptotic sector expansions nor proves a theorem with J growing with n . The sectors depend on the explicit cutoff convention; other sufficiently wide fixed relative windows change them by an exponentially smaller endpoint contribution. No canonical Borel sum, convergence of the infinite correction series, or exponential inverse theorem follows from the arguments above. The algebraic inverse of Part I (Theorem 7.1) remains valid independently.

Review status. The spectrum and real all-orders input was independently audited. Both new proof components received independent cross-review, followed by integrated mathematical review. These checks are not formal verification or conventional peer review. The computations supplement the proof and do not replace it.

[write, 2 October 2026] The three programs and their outputs are shipped as `code/56-fourier-*` and `data/56-fourier-*`; the coefficient generators that source 56’s replay

also calls are Part I's (`code/57-catalan-residue_expansion.py` and `code/57-catalan-saddle_expansion.py`), of which source 56 carried byte-identical copies. At intake (2 October 2026) `verify_fourier_sectors.py` and `verify_exponential_spectral_replacement.py` reproduced their delivered outputs; the order-192 refinement and the consistency check were not rerun (README).

References

- [1] OEIS Foundation, *A113227*, entry inspected 1 October 2026. Formal continued fraction credited to S. N. Gladkovskii, 10 October 2012. <https://oeis.org/A113227>.
- [2] S. Elizalde, *Asymptotic enumeration of permutations avoiding generalized patterns*, Advances in Applied Mathematics **36** (2006), 138–155. doi:10.1016/j.aam.2005.05.006; arXiv:math/0505254.
- [3] D. Callan, *A bijection to count (1-23-4)-avoiding permutations* (2010). arXiv:1008.2375.
- [4] N. R. Beaton, M. Bouvel, V. Guerrini and S. Rinaldi, *Enumerating five families of pattern-avoiding inversion sequences; and introducing the powered Catalan numbers*, Theoretical Computer Science **777** (2019), 69–92. doi:10.1016/j.tcs.2019.02.003; arXiv:1808.04114.
- [5] E. Duchi, V. Guerrini and S. Rinaldi, *A generating tree for permutations avoiding the pattern 122⁺3*, Fundamenta Informaticae **163** (2018), 21–39. doi:10.3233/FI-2018-1730.
- [6] Z. Lin and S. Fu, *On 120-avoiding inversion and ascent sequences*, European Journal of Combinatorics **93** (2021), 103282. doi:10.1016/j.ejc.2020.103282; arXiv:2003.11813.
- [7] M. Cervetti, *A generating tree with a single label for permutations avoiding the vincular pattern 1-32-4*, Pure Mathematics and Applications **30** (2022), 56–62. doi:10.2478/puma-2022-0009.
- [8] A. Frosini, V. Guerrini and S. Rinaldi, *Constrained underdiagonal paths and pattern avoiding permutations*, Mathematics **13** (2025), 517. doi:10.3390/math13030517.
- [9] B. Testart, *Generating trees growing on the left for pattern-avoiding inversion sequences*, Discrete Mathematics and Theoretical Computer Science **27:1** (2025), article 10. arXiv:2411.05726.
- [10] M. Pétréolle, A. D. Sokal and B.-X. Zhu, *Lattice paths and branched continued fractions: an infinite sequence of generalizations of the Stieltjes–Rogers and Thron–Rogers polynomials, with coefficientwise Hankel-total positivity*, version 2 (2020), Theorem 9.9. arXiv:1807.03271.
- [11] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*. Part I: Chapter 10, especially §§10.2, 10.4, 10.5 and formula 10.9.7, <https://dlmf.nist.gov/10>. Part II: §§4.13, 5.11, 10.9, in particular Schlöfli's integral <https://dlmf.nist.gov/10.9.E7>, complex Stirling estimates <https://dlmf.nist.gov/5.11>, and Lambert branches <https://dlmf.nist.gov/4.13>. [Merged from the two sources' entries.]
- [12] R. B. Paris, *The asymptotics of the Touchard polynomials*, 2016, <https://arxiv.org/abs/1606.07883>.